ITAI 1371

Module 13 – Contribution Journal

Student: Alexander DeBusk

Individual submission follows

L13 – Contributions – Alexander DeBusk

- Restored the original code formatting and organized the code into readable sections.
- Completed the reflective knowledge check by explaining the advantages of Pipelines, data leakage prevention, and how to extend a Pipeline with PCA.
- Reviewed the differences between manual preprocessing and the Pipeline approach.
- Completed the reflection document.
- Correctly saved all deliverables as PDFs using the appropriate file-naming conventions and submitted them to the group leader on time.

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Module 13 – Contribution Journal

Student: Shareefah Adeola

Individual submission follows

CONTRIBUTION JOURNAL: Module 13: Building Machine Learning Pipelines

Learning

ITAI 1371

Student: Shareefah Adeola

Date: November 13, 2026

I participated in the practical implementation of the Module 13 Machine Learning Pipeline exercise using Google Colab.

My contributions included:

1. Loading and examining the Titanic dataset using Python and Pandas.
2. Performing manual data preprocessing, including feature selection, scaling, and categorical encoding.
3. Splitting the dataset into training and testing sets.
4. Building and evaluating a Random Forest classification model using the manual approach.
5. Developing a Machine Learning Pipeline using make_pipeline and make_column_transformer.
6. Applying SimpleImputer, StandardScaler, and OneHotEncoder to the appropriate features.
7. Fitting the complete Pipeline, generating predictions, and evaluating model accuracy.
8. Identifying and resolving coding errors encountered during implementation.
9. Comparing the performance of the manual and Pipeline approaches.
10. Completing the reflective knowledge questions covering Pipeline advantages, data leakage, PCA, and real-world deployment.

I also reviewed the code, interpreted the results, and documented my learning outcomes.

The exercise strengthened my practical understanding of Python programming, data preprocessing, classification models, and Machine Learning Pipelines.

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Module 13 – Contribution Journal

Student: Yilin Leng

Individual submission follows

Contribution

In this Module 13 Lab, I used Google Colab and the Titanic dataset to complete two approaches: manual preprocessing and the Pipeline method. I handled missing values, separated numerical and categorical features, and used StandardScaler and OneHotEncoder to prepare the data.

After that I combined the preprocessing steps with a RandomForestClassifier using make_column_transformer and make_pipeline. After that, I trained the model, made predictions using the test data, and checked its accuracy with accuracy_score. I also answered the reflection questions about data leakage, PCA, and the benefits of using Pipelines.